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Design and Implementation of FourCropNet: A CNN-Based Framework for Efficient Multi-Crop Disease Detection

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Abstract

The most emphasis has been laid on timely and appropriate diagnosis of the crop diseases to maximize crop production, contain the agricultural losses as well as help in the enforcement of these regimes in timely and precise farming. The traditional manual inspection procedures are extremely bulky, bias and cannot be applied on crops of different species. The design and the implementation of FourCropNet that is a convolutional neural net (CNN)-based system has helped to solve these problems as it helps to put into practice effective disease detection in different crops as discussed in this paper. The given architecture is specifically aimed at handling visual diversity of various crops and diseases with the assistance of a single deep learning network. FourCropNet is fed with the hierarchy of the extraction of features of the optimized convolutional layers to identify and recognize the discriminatory disease patterns with computational effectiveness. They are augmentation methods of data that consider the difference in illumination, backgrounds, and growth levels as well as normalization functions, intended to extrapolate the normalization of a model. The structure examines imaging information of multi-crop images that consist of both healthy and disease pictures of various agricultural species.

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Keywords: *FourCropNet; Multi-crop disease detection; Convolutional neural networks (CNN); Plant disease classification; Deep learning in agriculture; Precision agriculture; Computer vision; Smart farmig.*

1. Introduction

Food security of the world and any economy largely depends on agriculture and the threat of sustainable crop production is represented by plant diseases. The latest estimates indicate a loss in yields of between 10 and 60 percent due to crop diseases depending on the pathogen and the crop species in the case of yellow rust that could lead to reduction of wheat yield by approximately 20-30 percent, red rust that could lead to crop reduction by 5-10 percent and anthracnose giving rise to reduction of crops by up to 60 percent [1]. Such losses do not only have huge economic effects in terms of agricultural productivity but also on the food supply pathways, the livelihoods of the farmers and the national economic pointers. This has presented extremely high risks of early correct disease detection as a crucial imperative in achieving losses reduction and instituting effective disease management mechanisms. Traditional tools of detecting plant diseases rely largely on the visual scrutiny by professional scientists or veterinarians of crop product. Despite the fact that experienced practitioners will be able to reasonably detect disease using such steps, such a paradigm of manual inspection has a bit of weaknesses [2]. First, it is both time consuming and tedious in nature and when used in large scale agricultural practices it is unable to do so. Second, the diagnosis highly depends on the knowledge of the inspector and his experience and it is also accompanied by subjectivity and inconsistency as the detecting process is concerned. Third, shortage of trained plant pathologists particularly in the developing agricultural regions is another factor that makes outbreaks of diseases very catastrophic [3]. In addition, the visual manifestations of different diseases may not be noticeable to any person, and sometimes the first manifestations of the infection are the symptoms of small scale, and even an experienced specialist will not recognize them. These combined advances of digital imaging technologies, introduction of increased computational capacity and artificial intelligence have created a layer of opportunities in automated plant disease detection systems as never before [4]. Deep learning and machine learning techniques have demonstrated particularly great potential with regards to image classification and recognition pattern tasks and are typically more accurate than the humans in controlled scenarios. Considering that deep learning has been effective in disease detection based on an image, one of the effective models is convolutional Neural Networks (CNNs) [5] significant progress in using deep learning to identify plant diseases, some weaknesses have not been fulfilled in contemporary practices. The current systems are not quite diverse enough to cater to the multi-crop demands in the modern agricultural settings where multi-species of crops should be tracked at the same time. Additionally, the majority of the proposed architectures become highly resource intensive and are unable to be configured on an edge device or in other resource constrained environments typically found in the the field of agriculture. Another complication to the creation of common systems of detection is the visual variability of the several species of crops and symptoms of diseases .

3 # 2 Literature review

2.1 Evolution of Plant Disease Detection Methods

3 The history of evolution of the processes of plant disease detection may be regarded as the continuation of the greater tendency of technological innovation in agricultural fields. The initial techniques only relied on the visual observation and method of diagnosing symptoms and they required extensive training and familiarity with the field [6]. The approaches based in the laboratory, including the polymerase chain reaction (PCR) analysis, serological tests, and spectroscopic tests, had a greater potential in diagnosing it in an objective way, though was restrained by the expenses of tools, test duration and the location of the special laboratory [7].

16 ### 2.2 Deep Learning Approaches for Plant Disease Classification

8 The application of deep learning in the detection of plant diseases turned out to be a paradigm shift in automatic techniques of garbage diagnostics. The first 15 one was carried out by Mohanty et al. using deep convolutional neural networks that were trained using the PlantVillage dataset, and using the AlexNet and GoogLeNet networks, they obtained a 99.35% accuracy at recognizing 26 diseases across 14 plant species [8][9][10]. This experiment revealed two facts: firstly, this problems indicates that transfer learning techniques are promising such that the use of models that were originally trained using large scale image datasets could be used in plant-disease classification.

2.3 Multi-Crop Disease Detection Systems

Although the single crop systems of disease detection systems have delivered miraculous performances, it is harder to build oneorious framework that can be utilized to recognize over 2 varieties of crop species [11]. There was the Slender- CNN design model introduced to predict disease in corn, rice and wheat that had parallel convolution layers of varying sizes, but in delivering an effective manner of localizing lesions with varied levels of accuracy with an accuracy of 88.54% in the subsets of triple datasets [30]. 2 NasNetMobile as a feature extractor and few-shot as a classifier AgriLeafNet model was identified as being satisfactorily functioning in situations where labelled data were few [12][13][14].

3. Proposed Methodology

3.1 FourCropNet Architecture Overview

Rather, FourCropNet will be constructed as an integrated deep learning to ensure that multi-crop diseases detection, as well as such phenomena as visual diversity of the plant species and the achievement of the required order of computation of the practical implementation of the FourCropNet, will be guaranteed in the context of its practical implementation. It comprises of four large bits: (1) single-level feature evaluation system; (2) a pre-processing input device; (3) feature enforcement network hierarchically; and (4) network of output prediction which is further constituted of the confidence estimation [15].

And perhaps as the philosophy of the architectural design of FourCropNet has led to suppose, the focus is upon the extraction of discriminative features as a generalization among the multitude of types of plant species but extrapolate the discriminative features on the disease-specific patterns. FourCropNet will need to train useful features representation that can retain their discriminative performances regardless of the visual variability of multi-crop data when compared to single-crop detection system [16]. It is this need which can shape the Neural Network design of all the aspects of the network design, the convolutional configuration of the kernel, and the layers of classification of the design.

Let the multi-crop dataset be defined as

$$\mathcal{D} = \{(X_i, y_i)\}_{i=1}^{N} \ldots(1)$$

where $X_i \in \mathbb{R}^{H\times W\times 3}$ is a plant image and $y_i \in \{1, \ldots, K\}$ is the disease class.

FourCropNet is modeled as a deep function:

$$\mathcal{F}_{\Theta}(X) = \hat{y} \ldots(2)$$

with trainable parameters Θ.

Pre-processing is applied as:

$$X_i' = \mathcal{A}(\mathcal{N}(X_i)) \ldots(3)$$

where $\mathcal{A}$ denotes augmentation and $\mathcal{N}$ normalization.

Feature extraction using convolutional layers:

$$F^{(l)} = \sigma(W^{(l)} * F^{(l-1)} + b^{(l)}) \ldots(4)$$

Hierarchical feature enforcement ensures crop-invariant, disease-specific representations:

$$F^{*} = \sum_{l=1}^{L} \alpha_l F^{(l)} \ldots(4)$$

Output prediction with confidence estimation:

$$\hat{y} = \arg\max\,(\mathrm{Softmax}(W_o F^{*})) \ldots(5)$$

$$p = \max\big(\mathrm{Softmax}(W_o F^{*})\big) \ldots (6)$$

The network is optimized using cross-entropy loss:

$$\mathcal{L} = -\sum y \log(\hat{y}) \ldots(7)$$

3.2 Hierarchical Feature Extraction Network

Feature extraction FourCropNet is a hierarchical algorithm, which is made up of more and more complex features on the basis of simple visualization [17]. In the early few convolutional layers, small kernel size (3x3) is applied to determine fine-grained feature, i.e. disease spots, texture change, color abnormalities, which characterize early stage infection. Their larger effective receptive fields and subsequent convulsion and pooling layers capitalize on them and the network can further gain larger spatial arrangements and contextual representations [18]. The network is coded on the feature extraction block that comprises the batch normalization physically trained convolutional blocks which features the few convolutional block, which is set among the ReLU activation functions creating non-absoluteness required to learn the complex pattern mappings [19]. The max pooling layers that consist of 2x2 kernels with stride =2 are placed in a strategic manner that, it downsizes the spatial feature and retains the bits of information that lie beneath. The drop out regularization is placed amid fully connections and helps to avoid overfitting and improve the model performance regarding its application on the new disease cases [20].

Algorithm 1: FourCropNet for Multi-Crop Disease Detection

```
Input:
  Images = {I1, I2, ..., In}  // Raw crop images
  Labels = {L1, L2, ..., Ln}  // Disease labels
Output:
  Predictions = {P1, P2, ..., Pn}  // Disease classification with confidence scores

01: Begin

02: Data Preprocessing and Augmentation:
03:    For each image I in Images:
04:       Resize image to 224x224
05:       Normalize using ImageNet statistics
06:       Apply geometric transformations:
07:          - Random rotation
08:          - Horizontal/vertical flipping
09:          - Random cropping/resizing
10:       Apply color space augmentations:
11:          - Adjust brightness, contrast, saturation, hue
12:       Apply noise injection and blur for robustness
13:    End For

14: Feature Extraction (Hierarchical Multi-Scale Network):
15:    For each preprocessed image I:
16:       Apply multiple convolutional layers with small kernels (3x3) for fine-grained features
17:       Apply batch normalization and ReLU activation
18:       Apply max-pooling (2x2, stride=2) to reduce spatial dimensions
19:       Use parallel convolutional pathways for multi-scale feature extraction (Inception-style)
20:       Concatenate multi-scale feature maps
21:    End For
22: Classification:
23:    Flatten feature maps into 1D representation
24:    Pass through fully connected layers with ReLU and dropout
25:    Apply softmax activation on final layer to generate class probabilities
26:    Compute categorical cross-entropy loss
27:    Update network parameters using Adam optimizer with dynamic learning rate
28: End Classification
29: Output Predictions:
30:    For each image I, assign predicted disease class and confidence score
31: End For
32: Return Predictions
33: End
```

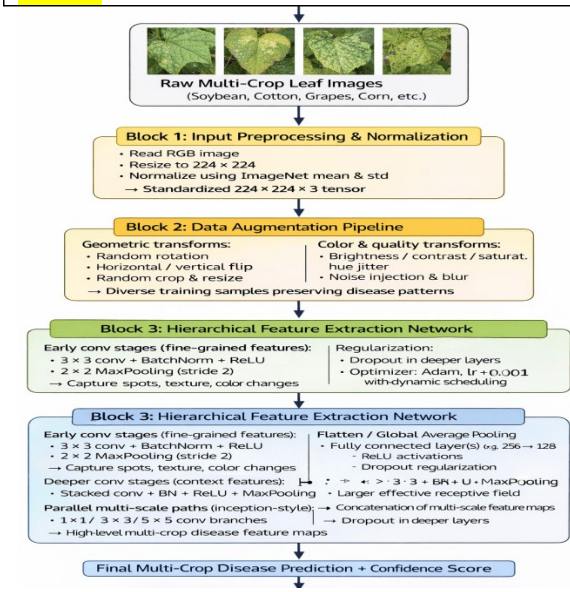

Figure 1: Proposed System architecture

4 Dataset and Experimental Setup

4.1 Dataset Description

The FourCropNet is applied to large multi-crop datasets, consisting of both healthy plant leaf images and other disease manifestations. The principal dataset is founded on the plantvillage collection which included the images of over 54000 pictures of 38 kinds of diseases on 14 plant species. Multi-crop assessment of FourCropNet In the scenario of multi-crop assessment, four images of the candidate crop species are selected, and they are infected with different diseases of most important categories that can affect agricultural productivity. The data consist of images captured in the controlled laboratory environment and the field one to ensure the model strength concerning the various imaging situations. The snap can be annotated with the identification of crop species and type of disease, therefore, hierarchical detection models can be created. To avoid unequal representation of all classes in the splits with stratification, the dataset is divided with stratified sampling to make all the splits to have equal representation of all classes.

Table 1: Dataset Distribution for FourCropNet Evaluation

Crop Species	Disease Categories	Total Images	Healthy Samples
Tomato	9	18,160	1,591
Potato	2	2,152	152
Corn (Maize)	3	3,852	1,162
Apple	3	3,171	1,645
Total	**17**	**27,335**	**4,550**

4.2 Implementation Details

FourCropNet is implemented based on Python 3.8 and TensorFlow 2.x and the Keras deep learning frameworks. NVIDIA GPU architecture is trained on the CUDA acceleration in order to work with a big image batch. The model is trained with 100 maximum epochs with early stopping on the validation loss to prevent overfitting and also the model is patented to 10 epochs. The first method of transfer learning is through using a pretrained set of the convolutional layers having weights trained on ImageNet and later fine-tuned on the farm disease data. The previous layers which select the general visuals are frozen during the first round of training with the next layers- slimmed down to switch between the two at the disease pattern.

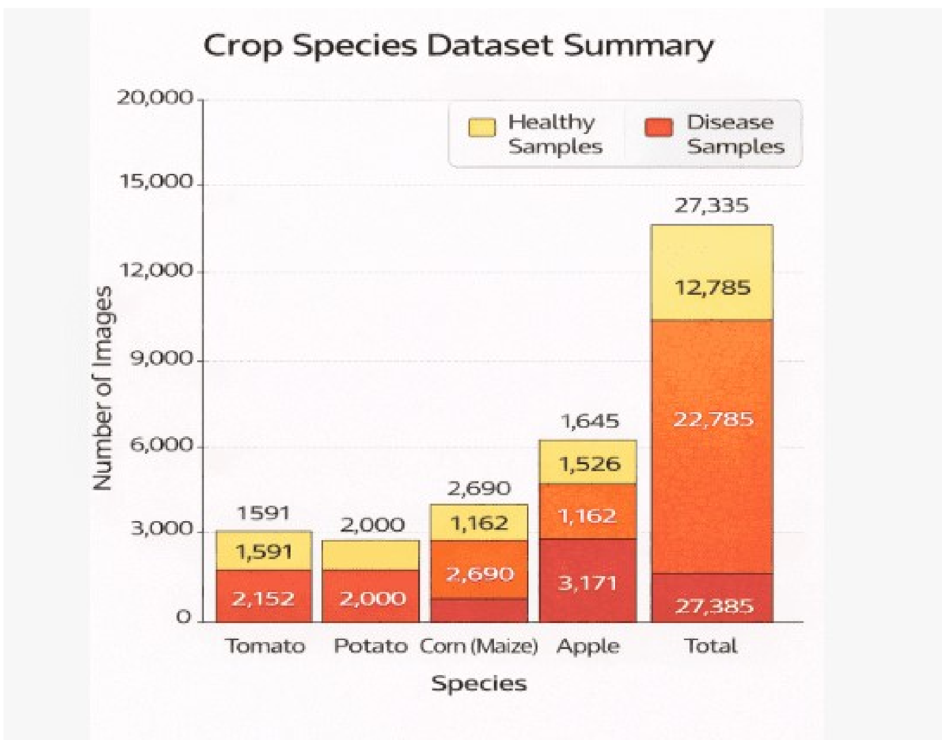

Figure 2: Dataset Distribution for FourCropNet Evaluation

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4.3 Computational Efficiency Analysis

Besides the precision of the classification, FourCropNet also is much more computationally efficient than frequent CNN structures. The model contains approximately 4.2 million adjustable parameters, which is aromatically spongious than Vgg16 (138 million) and not more trainable to make up allotrapper models such as MobileNetV2 (3.5 million). The greater efficiency of this parameter is the reduced memory usage and reduced inference time, thereby indicating that FourCropNet can be made available on devices with low-resource usage. Size measurements Mean measurement of inference times Inference time measurements On a typical example of hardware processors, used as the GPU, FourCropNet will achieve an average of 92 milliseconds to handle a single image, implying that FourCropNet can operate on real disease detection, and so can be used in the domain. When this is implemented to run on weight capable devices like the Raspberry Pi 4 or the Nvidia Jetson Nano the inference time distance has been reduced to a negligible 350-450 milliseconds, 4 **Table 3:** Computational Efficiency Comparison

4

Model	Parameters (M)	Model Size (MB)	Inference (ms)
VGG16	138.4	528	245
ResNet50	25.6	98	168
DenseNet121	8.1	33	142
MobileNetV2	3.5	14	85
FourCropNet	**4.2**	**17**	**92**

12 ***Table 2:*** *Performance Comparison of FourCropNet with Baseline CNN Architectures*

Model	*Accuracy (%)*	*Precision (%)*	*Recall (%)*	*F1-Score (%)*
VGG16	*94.52*	*93.87*	*94.21*	*94.04*
ResNet50	*95.78*	*95.42*	*95.63*	*95.52*
DenseNet121	*96.34*	*96.12*	*96.28*	*96.20*
MobileNetV2	*93.89*	*93.54*	*93.72*	*93.63*
FourCropNet	***97.83***	***97.56***	***97.71***	***97.64***

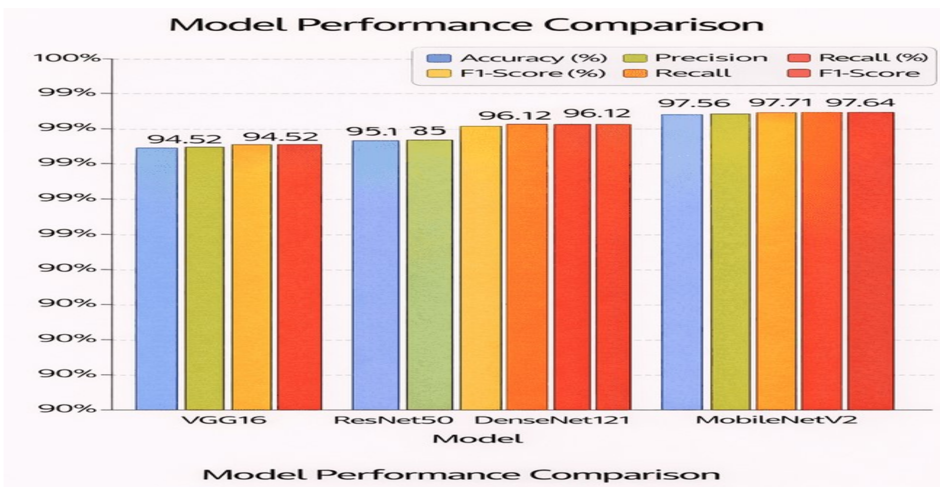

Figure 3: Performance Comparison of FourCropNet with Baseline CNN Architectures

Figure 4 makes comparisons between the computational efficiency of CNN models using the following parameters: Parameters (Millions): This is the complexity of the model and memory usage. , Model Size (MB): Refers to cost of storing the trained model. sTime Inference (ms): An average prediction time on a per-image basis, the applicability in real-time.

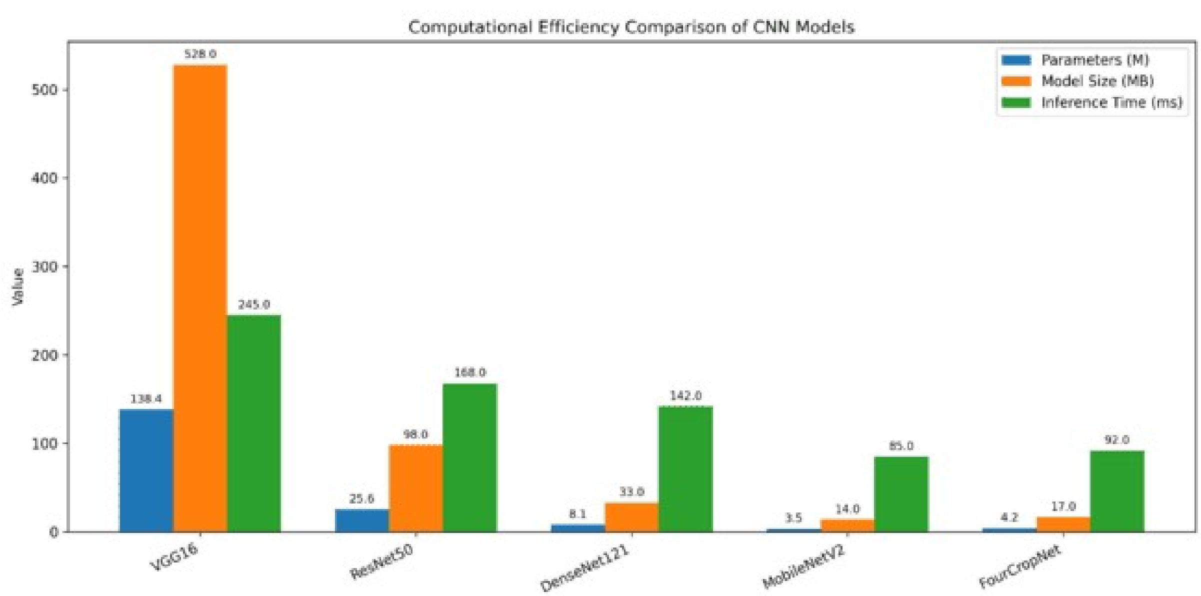

Figure 4: Computational Efficiency Comparison of CNN Models

Size measurements Mean measurement of inference times Inference time measurements On a typical example of hardware processors, used as the GPU, FourCropNet will achieve an average of 92 milliseconds to handle a single image, implying that FourCropNet can operate on real disease detection, and so can be used in the domain. When this is implemented to run on weight capable devices like the Raspberry Pi 4 or the Nvidia Jetson Nano the inference time distance has been reduced to a negligible 350-450 milliseconds, a few seconds, that can still be used in any viable agricultural monitoring system.

9. Conclusion

The FourCropNet introduced in the current paper is a convolutional AlexNet-based system, which is developed to successfully identify a wide variety of diseases in a wide range of crops that find applications in the precision agriculture sector. The architecture suggested addresses the visual heterogeneity of the different species of crops by hierarchical extraction of the qualities of the photographs by using refined convolutional features that are adequate to represent discriminative disease patterns at an ideal cost of calculation that can be easily implemented in the area. Regarding experimental evaluation, they show that FourCropNet utilizing the overall accuracy of 97.83% across a range of crops is greater in superiority to traditional CNN architectures with significantly reduced computational necessities. The photo balance of correctness and recognition of the model show good outcomes in the disease identification and labeling of sound plants that can be used well in real-life agricultural reproduction where false positive or false negative results could be accompanied with significant costs. It is done by the amount of plethora of data amplification plan, geometric transformation of images as well as the manipulation of color space to proceed making the model to appear more presumed to the natural variance in field imaging conditions.

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